

**MACHINE LEARNING**

**Questions to be set:** 05 (All Compulsory)

**Objectives:** This course aims to introduce students to foundational machine learning techniques and algorithms. It focuses on building the ability to select, apply, and evaluate appropriate models such as decision trees, neural networks, Bayesian classifiers, and genetic algorithms for solving real-world problems through data-driven learning and intelligent decision-making.

**Pre-requisites:** Fundamentals of probability & statistics and programming skills, data mining concepts.

**Course Outcomes (CO):**

|     |                                                                                                                 |
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| CO1 | Understand core concepts of supervised, unsupervised, and evolutionary learning techniques in machine learning. |
| CO2 | Apply algorithms such as decision trees, k-NN, neural networks, and clustering to basic learning tasks.         |
| CO3 | Analyze the strengths, limitations, and evaluation metrics of various machine learning models.                  |
| CO4 | Design simple machine learning workflows using suitable models and data preprocessing strategies.               |
| CO5 | Compare multiple learning approaches and interpret their applicability to real-world problems.                  |

| Module                                                                                | Topics to be covered | Topics                                                                                                                                                                                                                                                                                                                                                        | Hrs. |
|---------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Module 1:<br>Foundations<br>of Machine<br>Learning                                    | in<br>class          | Introduction to Machine Learning: Definition, Types (Supervised, Unsupervised, Reinforcement), Real-world Applications, Concept Learning and Hypothesis Space, Model Selection and Generalization, Overfitting, Underfitting, and the Bias-Variance Tradeoff, Evaluation Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, Cross-validation Techniques | [9]  |
|                                                                                       | Assignment<br>Topics | To be provided by the concern faculty members.                                                                                                                                                                                                                                                                                                                |      |
| Module 2:<br>Supervised<br>Learning I –<br>Decision<br>Trees &<br>Rule-Based<br>Model | in<br>class          | Decision Tree Learning: ID3, C4.5, CART, Entropy, Information Gain, Gini Index, Pruning Techniques (Pre & Post Pruning), Rule-based Learning: IF-THEN Rules, Sequential Covering, FOIL Algorithm, Applications and limitations                                                                                                                                | [9]  |
|                                                                                       | Assignment<br>Topics | To be provided by the concern faculty members.                                                                                                                                                                                                                                                                                                                |      |
| Module 3:<br>Supervised                                                               | in<br>class          | Artificial Neural Networks: Perceptron and Multilayer Feedforward Networks, Activation                                                                                                                                                                                                                                                                        | [8]  |

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|------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Learning II – Neural Networks & Bayesian Learning                |                   | Functions (Sigmoid, ReLU, Tanh), Training with Backpropagation, Introduction to Deep Learning, Bayesian Learning: Naive Bayes Classifier, Bayesian Belief Networks, Probabilistic Inference                                                                                                                                                                                               |      |
|                                                                  | Assignment Topics | To be provided by the concern faculty members.                                                                                                                                                                                                                                                                                                                                            |      |
| Module 4: Instance-Based Learning and Ensemble Methods           | in class          | k-Nearest Neighbours (k-NN), Case-Based Reasoning, Locally Weighted Regression, Lazy vs. Eager Learning, Ensemble Learning: Bagging, Boosting, Random Forest, Model Comparison and Performance Evaluation                                                                                                                                                                                 | [8]  |
|                                                                  | Assignment Topics | To be provided by the concern faculty members.                                                                                                                                                                                                                                                                                                                                            |      |
| Module 5: Unsupervised Learning, Evolutionary & Advanced Methods | in class          | Introduction to Unsupervised Learning: Clustering vs. Association, K-Means and Hierarchical Clustering, Dimensionality Reduction: PCA, Introduction to Evolutionary Algorithms: Genetic Algorithms (GA), Operators: Selection, Crossover, Mutation, Fitness Functions and Elitism, Applications of GAs in Learning, Brief Overview: Reinforcement Learning & Ethical Considerations in ML | [10] |
|                                                                  | Assignment Topics | To be provided by the concern faculty members.                                                                                                                                                                                                                                                                                                                                            |      |

#### **Textbooks:**

1. Tom M. Mitchell, Machine Learning, McGraw-Hill Publishers
2. Ethem Alpaydın, Introduction to Machine Learning, The MIT Press Cambridge, Massachusetts, London, England

#### **Reference Books:**

1. Ian H. Witten, Eibe Frank, Data Mining Practical Machine Learning Tools and Techniques with Java Implementations, Morgan Kaufmann Publishers.
2. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective (Adaptive Computation and Machine Learning series), MIT Press.